

1RT730 Project Report

LLM-Powered Career Assistant

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September 15, 2026

1 Introduction

Applying to a job or advancing your career is a complex task and can be overwhelming to new students or graduates who do not have the experience. One can easily get confused with the variety of tasks to be simultaneously done, such as searching for the right jobs, drafting letters and resumes, deciding what to upskill on and how, and several other tasks which are crucial for building your overall credibility for the job market. Even right now, as we are studying in this course, several students are looking simultaneously for Master's thesis and full-time opportunities. Our project is a **LLM-powered job seeking and career advancement assistant** that aims to guide users (graduates, early-career professionals, and job seekers) in the right direction, whether it's searching for the right jobs or hitting the right keywords on your resume. Our project is a multi-agent program that helps users in the following:

- Brainstorm about career guidance and upskilling
- Find open jobs tailored to their goals and skills
- Save searched jobs in user's calendar
- Draft Cover Letters based on well-researched templates

Note: GitHub Link for Project Given in References[\[2\]](#)

2 Architecture

The core skeleton of our platform is a FastAPI backend for core business logic, LLM orchestration, and Retrieval-Augmented Generation (RAG) services, and Gradio for front-end user interaction. The design prioritizes modularity and high-value workflow specialization by implementing a TF-IDF RAG system.

The platform provides three action paths: **General QnA**, a multi-step **Career Coach** (`/coach`), and a **RAG-grounded Cover Letter Generator** (`/cover-letter`).

2.1 Backend (FastAPI)

Entry Point: `app/main.py`

Function: handles request validation via Pydantic Data Models (`app/models/schemas.py`), managing all RAG operations, and executing all the specific action paths.

Routing: Specialized routers for Chat, Coaching, RAG maintenance, and Calendar Authorization.

Technology Stack: FastAPI, Uvicorn, Python-Dotenv, Pydantic, Requests.

2.2 Frontend (Gradio)

Entry Point: `ui.py`

Function: Multi-tabbed interface (Q&A, Coach, Upload, Cover Letter). Handles user input and file uploads before making API calls to the FastAPI backend.

2.3 Action paths

The application’s functionality is delivered through specialized action paths which execute as follows:

- **General QnA:** Handles quick, general queries about career, skill building and application help; Calls the `simple_chat` wrapper with a standard system prompt (We use specialized prompts to guide users in the right direction) and the user’s query (See Figure 2).
- **Career Coach Workflow:** Orchestrates the multi-tool `coach_flow()`; The LLM drafts improvements based on uploaded resume text, and specialized `search_jobs_*` tool uses Tavily to query multiple trusted ATS/job boards, filtering and ranking results (See Figure 3). Google Calendar integration schedules the different jobs into the user’s Google Calendar accessed through user-specific Google OAuth credentials (See Figure 4).
- **Cover Letter Generation:** Builds query from `user_profile`, `jd_text`, and optional resume. Stores the resume (if uploaded) in the RAG store. Searches RAG store for context (Established cover letter templates, uploaded resume). `generate_cover_letter` chain uses `PROMPT_TEMPLATE` and inserts retrieved context to draft a tailored letter for further polishing and use (See Figure 1).

Additionally, our Career Assistant uses a RAG layer built for uploading cover letter templates and resumes. All RAG data is stored locally for simplicity, with original uploaded files placed in the `uploads/` directory and the primary index maintained in `rag_store/meta.jsonl`, where each JSON object represents a text chunk. The RAG layer provides maintenance endpoints for operational management (`/rag/stats` for reports metrics - total number of text chunks, sources, and storage usage; `/rag/clear` to delete the `meta.jsonl` index, with an optional flag to clear `uploads/` directory).

2.4 Security and Future-Proofing

- For our Google Calendar integration, our system uses Google OAuth tokens, which are stored per user at `./credentials/tokens/user_id.json` to enable authenticated calendar operations.
- For Data Validation, File size/type checks are done on ingestion endpoints.
- The system keeps a consistent `search(query, k)` signature to allow future migration to vector databases like Chroma or FAISS.
- Also, the `simple_chat` function accepts system overrides for fine-grained persona control in specialized workflows.

3 Dataset

Since the application of job seeking and career advancement is a highly personalized one, a fixed dataset was not used. Instead, our data sources were synthesized by applying RAG that inputs and indexes various documents such as cover letter templates and resumes that the user uploads through the Upload path. For example, we tested the cover letter generation path by uploading the famous U.Wisc Madison Cover Letter Writing Template[1] to polish the draft generation. The type of data typical to this application are PDFs, such as cover letters and resumes.

During upload, documents are parsed using `pypdf` and `python-docx`, and the text is split into chunks using a sliding window approach (maximum 900 characters per chunk with 150-character overlap) to preserve context across boundaries. Retrieval is based on TF-IDF embeddings and similarity calculations implemented with `scikit-learn` for effective keyword-based matching for resumes, job descriptions, and cover letter templates.

4 Results

The outcomes of our project is presented in terms of qualitative and quantitative results.

4.1 Metrics for cover letter

For the cover letter endpoint, we are using a file `metrics.py`. It is a batch evaluation tool which automates testing multiple cases, measuring latency, and computing text quality metrics to benchmark the system.

The inputs are:

- CSV file containing test cases with headers: `user_profile`, `jd_text`, `target_company`, `target_role`, `style`, `resume_path`.
- Optional resume file uploaded if `resume_path` exists.
- CLI arguments: `--api` (base URL), `--csv` (input CSV), `--out` (CSV output), `--summary` (Markdown summary output).

The program sends `GET /health` to verify API availability, reads each test case row, posts text fields and (optional) resume to `/cover-letter`, measuring end-to-end latency, extracts generated letter, used context, and retrieval scores from JSON. The metrics are:

- JD similarity (TF-IDF cosine) between generated letter and JD.
- Paragraph count, simple greeting/closing presence check using RE and pattern matching, word and character counts, average sentence length.
- Context usage: number of snippets used, average and top retrieval scores.

The program outputs a CSV file with these metrics and other diagnostic fields, and a markdown summary aggregating success rate, average/percentile latency, JD similarity, and context statistics.

4.2 Quantitative results for cover letter

For simple test case:

Table 1: Evaluation Dataset for Cover Letter Generation

User Profile	Job Description Text	Target Company	Target Role	Style
MSc DS; Python, SQL; RL & control projects	Thesis: Nonlinear MPC...	Zeekr	Master's Thesis (NMPC)	data-driven
MSc CS; AI engineer intern	Masters Thesis in Multimodal Knowledge-Extraction of Clinical Data using LLMs	KTH University	LLM Research	concise
Masters in Molecular Medicine; Honors in Biomedical Sciences	Thesis work, 30 credits - Advancing Bioanalytical Techniques for Drug Discovery	AstraZeneca	Medicine Research	formal

We get the following results:

- JD Similarity = **0.077 — 0.143**
- Paragraphs = **7**
- Average Latency = **2759.2 ms**
- Average context count = **6**

The JD similarity is low considering that it is a simple snippet used for testing. Upon using a more detailed snippet (copying an example JD from AstraZeneca Careers Site), the JD similarity goes up to **0.2 — 0.3**. The paragraph count includes greeting and closing paras, so average paragraph count is **5**, which satisfies the prompt.

4.3 Other Qualitative results

- The generated cover letter is simple and concise, as it maps the JD requirements to the skills of the user as indicated in his resume. With the cover letter templates uploaded to the RAG Store, the generated cover letter also follows a template with the recommended structure and style and avoids invented claims.
- The retrieved snippets are helpful with JD similarities around the 0.2 — 0.3 mark (as mentioned before) provided a fairly detailed JD is specified.
- The application is highly personalized with guardrails (constrained prompts, RAG, job search filters), to ensure it gives relevant points regarding career advice without deviations.
- The coach provides actionable results by storing the extracted job details in the user's calendar.

4.4 Limitations

Our project, like any other, suffers from these limitations.

- We don't share a reusable context plan across flows. A small orchestration layer for consistent keywording, snippet selection, and structure could be added.

- We could optimize the use of LLM Calls via caching, prompt optimization, and compression techniques.
- The current version uses Langchain for RAG and is great for understanding fundamentals but could be improved for productivity by using an agent framework could automate planning, retries, tool use, and guardrails.
- We could add AI reranking/summarization for postings, with caching to control extra LLM calls.
- We could use MCP servers (LinkedIn, Calendar, etc.) to parse and execute actions with minimal code.

5 Societal Impact

Deploying an LLM-powered job-seeking and career advancement assistant offers meaningful societal upside, but also brings serious responsibilities. On the positive side, it can lower barriers to opportunity. It gives first-time applicants and underrepresented groups instant access to high-quality guidance, which previously required costly coaching. It helps users discover unknown roles, translate skills into market language, quickly iterate resumes and cover letters, and learn which competencies to upskill. These benefits improve efficiency and democratize career knowledge at scale. It can also enhance education by making job search into a formative, feedback-rich process. Users learn how to scope roles, analyze requirements, and reflect on their strengths and gaps.

Yet, risks are real. The model can replicate and amplify historical biases in labor data, such as gendered or racialized patterns in job ads, and could unfairly rank candidates. It may hallucinate or oversell credentials and surface misleading "fit" suggestions. Uploading, scraping, or processing user data raises privacy and consent concerns. Sensitive information like names, surnames, employment history, or disability disclosures can be exposed and possibly re-identified if combined with public sources. Retrieving certain job site information may also violate terms of service, copyright, or robots.txt.

Our system can automatically log model-generated job search notes to Google Calendar after generating recommendations. This feature could be misused to schedule or legitimize spammy or deceptive applications. The intensive model inference required produces non-trivial environmental costs. It consumes significant compute energy and creates carbon emissions, since large embeddings and retrieval layers increase memory and IO overhead. Frequent fine-tuning and vector re-indexing cause additional compute spikes. Data centers may also have high water usage for cooling.

To mitigate these risks, we practice data minimization and collect only what is needed. We use local or encrypted storage, allow for granular consent controls, provide user-visible data logs, and offer deletion options. We adopt explicit fairness goals, such as parity checks across protected attributes where legally permissible. Continuous bias monitoring with measurable thresholds and rollback plans is in place. The system can calibrate confidence, cite sources for claims about roles or skills, and label generated text to support transparency and contestability. Safety mechanisms include guardrails against discriminatory or exploitative advice and filters for sensitive attributes. We also report energy use where feasible, choose efficient architectures, and schedule heavy jobs responsibly.

Taken together, these practices acknowledge that LLMs can broaden access and accelerate learning in the labor market. However, responsible deployment requires privacy by design and transparent reasoning.

Use of generative AI The use of Generative AI was as follows:

- Generating simple boilerplates for the utility tasks in code.
- Rephrasing and polishing text in Report (with Grammarly)
- Generating Interesting-looking PPT Designs

References

- [1] University of Wisconsin–Madison Writing Center. Writing cover letters. <https://writing.wisc.edu/handbook/coverletters/>, 2025. Accessed: 2025-10-19.
- [2] Abhinav Ramalingam, Battu Anudeep, and Tebogo Sanelo Mitane. LLMPProject: Career Coach & RAG Cover-Letter Assistant. <https://github.com/Abhinav-Ramalingam/LLMPProject/>, 2025. GitHub repository.

A Figures

Chat Coach Upload **Cover Letter**

Your profile / summary

Masters in Molecular Medicine; Honors in Biomedical Sciences

Job description (paste)

Thesis work, 30 credits - Advancing Bioanalytical Techniques for Drug Discovery

As a master's thesis student, you will gain comprehensive experience in bioanalytical techniques such as immunoassays and PCR. You will work with a diverse range of molecular modalities including proteins, antisense oligonucleotides (ASOs), mRNA, and siRNA, and become skilled in both method development and assay performance evaluation. Within our dynamic department, we continuously strive to innovate, seeking new ways to improve efficiency, sensitivity, and insight into our workflows.

Depending on your interests, you may explore topics such as:

Multi-endpoint analysis – evaluating whether a single biological sample can yield multiple endpoints effectively ("more with less")

Drafted Cover Letter

Dear Hiring Manager,

I am writing to express my strong interest in the Master's Thesis Student position in Molecular Biology at AstraZeneca. As a Master's student in Molecular Medicine with honors in Biomedical Sciences, I am eager to apply my knowledge of bioanalytical techniques to advance drug discovery.

My background includes experience with DNA and RNA isolation, PCR-RFLP, and cell culture, which align well with the requirements of this thesis. I am particularly interested in the opportunity to work with diverse molecular modalities, including proteins, ASOs, mRNA, and siRNA, and to contribute to innovations in efficiency and sensitivity within your department's workflows.

My previous research experience has equipped me with skills in method development and assay performance evaluation. I am excited by the prospect of tailoring my thesis to explore topics such as multi-endpoint analysis or workflow acceleration, and to make a meaningful contribution to the field.

Figure 1: Action Path: Cover Letter Generation

Chat **Coach** Upload Cover Letter

Use the general assistant endpoint /chat for quick Q&A or guidance.

Question / Prompt

what are promising research areas for future

Optional user profile

MSc Computer Science

Ask

Response

Based on your background in Computer Science, here are some promising research areas for the future:

- * **Artificial Intelligence (AI) and Machine Learning (ML):***
- * Explainable AI (XAI)
- * Federated Learning
- * Generative AI and foundation models
- * Reinforcement Learning
- * **Cybersecurity:***
- * AI-driven security
- * Quantum-resistant cryptography
- * Privacy-enhancing technologies
- * **Data Science and Engineering:***

Figure 2: Action Path: General QnA

LLM-powered Career Assistant

Chat

Coach

Upload

Cover Letter

Upload your resume (PDF/Docx/TXT)

SS_CV'25.pdf169.5 KB

Your profile / summary

Masters in molecular medicine

Job query

masters thesis in molecular biology

Location

sweden

Calendar (optional)

Create 'Apply today' event

Start (HH:MM)

End (HH:MM)

18:00

20:00

Okay, here's a strict resume review based on the information you provided.

Top Issues:

- Lack of Quantification:

The resume is missing quantifiable results and impact. Every experience needs to showcase what the candidate achieved, not just what they did.
- Fluff and Redundancy:

The "Profile" section is generic and adds little value. Phrases like "wide range of interests" and "make an impact" are vague and should be removed. Soft skills should be embedded within the descriptions of experiences and projects.
- Inconsistent Formatting and Dates:

Date formats are inconsistent, and some sections lack clear chronological order.
- Weak Action Verbs:

Many bullet points start with weak verbs like "Learned". Stronger action verbs would emphasize the candidate's initiative and contributions.
- Unclear Role Definition:

The "EXPERIENCES" section is too broad and lacks context. Each experience needs to be clearly defined (e.g., internship, research assistant).
- Missing Context on Guest Writer Role:

Details about the guest writer role need to be clear (e.g., what was written about and for what audience).

Rewrites for 3-5 Bullets:

- Original:

"Maintenance of mammalian cell culture and zebrafish larvae"

 - Rewrite:

"Maintained mammalian cell cultures (e.g., HeLa, HEK293) and zebrafish larvae, ensuring optimal growth conditions for [specific experiment type, e.g., drug screening] and contributing to [project goal, e.g., successful validation of 3 novel therapeutic targets]" (Need to add: Specific cell lines, purpose of maintenance, and impact)
- Original:

"An association study between miRNA-146a polymorphism and Autism Spectrum Disorder (ASD)"

 - Rewrite:

"Conducted an association study of miRNA-146a polymorphism and Autism Spectrum Disorder (ASD) using DNA extraction, PCR-RFLP, and SNP genotyping, resulting in [quantifiable result, e.g., identification of a statistically significant correlation between miRNA-146a polymorphism and ASD risk (p < 0.05)]" (Need to add: specific statistical results)

Figure 3: Action Path: Career Coach



Apply for jobs (today)

Today • 12:52–13:52



Invite via link



2 hours before

1 day before



Query: masters thesis in molecular biology

Location: sweden

Jobs to apply:

1. Science Careers Molecular Biology jobs in Sweden - Science Careers

<https://jobs.sciencecareers.org/jobs/molecular-biology/sweden/>

Molecular Biology jobs in Sweden . 1 job to view and apply for now with Science Careers

2. ResearchGate Best Molecular